Project Title: Predicting Credit Card Default Risk Using Data Mining Techniques

Member	Student Name	Student ID
1	Arvind a/l Gopalakrishna Thevar	1211104071
2	Santoshkumar a/l K Munusamy	241UC240LW

Abstract

This study handles the crucial problem of predicting credit card default risk, which is an essential part of financial risk management in the banking industry. Financial organizations can reduce losses and improve credit approval procedures by using precise predictions. This work uses binary classification to forecast the probability of missed payments using a dataset from the UCI Machine Learning Repository that includes the financial and demographic information of 30,000 Taiwanese clients.

The methodology involves thorough data preprocessing that addresses feature scaling and class imbalances before doing exploratory data analysis to pinpoint important risk indicators. Several machine learning models are created and contrasted, such as Random Forest, Logistic Regression, and Neural Networks. Metrics like ROC-AUC, F1-score, and accuracy are used to thoroughly assess the model's performance.

A dashboard prototype that allows bank officers and risk managers to see default probability and risk levels (low, medium, and high) based on client inputs is part of the project in addition to model development. In order to capture complicated consumer behaviours, ensemble approaches and correcting data imbalance are crucial, according to preliminary results and associated research. A strong predictive model that not only detects high-risk customers but also offers practical insights into the historical and demographic variables influencing credit default is the expected outcome.

1.Introduction

A key component of the contemporary financial sector is the control of credit risk, especially as credit card use continues to rise worldwide. Differentiating between dependable borrowers and those who are at danger of default is a persistent difficulty for financial organizations. In order to deliver data-driven insights that can help banks minimize financial losses and optimize their credit approval methods, this project focuses on Predicting Credit Card Default Risk.

Background and Motivation

A number of demographic and past financial characteristics influence the difficult challenge of predicting whether a customer would miss a payment. This study is motivated by the high stakes associated with financial risk assessment even a slight increase in forecast accuracy can result in large capital savings for institutions. Furthermore, class imbalances in real-world financial data, where non-default situations far outweigh default cases that make advanced data mining approaches necessary to guarantee the model's continued efficacy for high-risk identification. This study aims to close the gap between unprocessed financial data and useful decision support by utilizing the dataset of 30,000 Taiwanese credit card customers from the UCI Machine Learning Repository.

Objectives

The primary goal of this research is to develop a robust binary classification framework to forecast default risk. To achieve this, the project outlines the following specific objectives:

- To gain insight into and understand the financial and demographic factors that influence credit card default.
- To build and assess different classification models, such as Neural Networks, Decision Trees, and Logistic Regression.
- To evaluate the model's performance using strict criteria like F1-score, ROC-AUC, accuracy, precision, and recall.
- To create a useful application mockup that shows how financial risk managers and bank officers might use these predicted insights in real-time situations.

2.Related Work

The development of financial risk assessment through sophisticated data mining and machine learning approaches is highlighted in the research currently available on credit card default prediction. The particular problem of imbalanced datasets in this field was examined by Alam et al. (2020), who pointed out that real-world financial data usually contains considerably fewer default scenarios than non-default ones. Their study showed that enhancing the prediction accuracy of machine learning models requires resolving this class imbalance, which is also evident in the UCI Taiwan dataset. In the same way, Arora et al. (2021) investigated a number of techniques for forecasting defaults, such as logistic regression, decision trees, and ensemble methods. Their results

demonstrated that these models, which offer a methodological roadmap for feature selection and preprocessing, may successfully capture the complex trends seen in consumer financial behaviour.

Additional comparison analysis by Abbas and Hussein (2024) highlighted the significance of assessing several categorization methods, such as Random Forest and Neural Networks, instead of depending on a single model to evaluate bank customer credit rating. In addition to these algorithmic research, Ali et al. (2024) demonstrated how these prediction tools are essential decision support systems for financial institutions by presenting a data-mining-based model tailored for funding requests. Collectively, these studies demonstrate that a thorough comparison of different mining approaches and intelligent handling of data imbalances are necessary for accurate risk assessment.

Similarities and Differences:

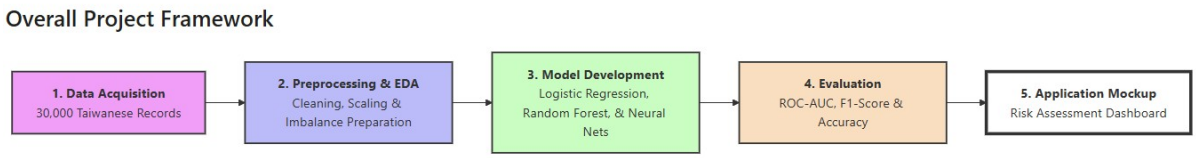
This project's goal of integrating demographic and historical financial data to solve a binary classification problem is one of the main commonalities with the evaluated literature. This study uses the dataset from the UCI Machine Learning Repository and concentrates on crucial preprocessing processes such feature scaling and handling class imbalances, much as the studies by Alam et al. and Arora et al. Additionally, the process of comparing several models such as Random Forest, Decision Trees, Logistic Regression, and Neural Networks in particular directly corresponds with the comparison framework that Abbas and Hussein support.

The project's focus on the real-world implementation of these discoveries through an application prototype is the main distinction. This project expands those findings into a useful dashboard intended for bank officers and risk managers, whereas the relevant research publications mostly concentrate on theoretical model performance and statistical validation. This study transcends the typical academic scope of the previous works by bridging the gap between sophisticated data mining outputs and actionable decision assistance through the use of a probability score and real-time visual indicators of risk (low, medium, and high).

3.Methodology

The methodology for this project is designed to provide a systematic approach to predicting credit card default, moving from raw data to a functional decision-support tool.

3.1 Overall Framework



The project uses a conventional data mining pipeline designed with binary classification in mind. Data collection and preliminary investigation are the first steps in the process, which is then followed by thorough preparation to guarantee data quality. Several machine learning models are trained and optimized after the data is ready. The next steps include creating an application mockup to show

practical utility and evaluating performance using a variety of metrics. This methodical methodology guarantees that the project's goals of identifying risk factors and developing a strong predictive model are fulfilled with scientific integrity.

3.2 Dataset

The research uses the Default of Credit Card Clients dataset which serves as a standard benchmark for data mining and financial risk analytics applications. The dataset contains 30,000 entries which represent individual credit card holders in each observation. The dataset contains 25 attributes which include one identifier column and 23 explanatory features and one binary target variable that indicates default behavior.

The dataset includes demographic data and financial behavior information which allows researchers to perform complete customer credit risk evaluations. The demographic variables which include gender and education level and marital status and age help organizations understand their customers' social economic characteristics. The financial variables in this system include credit exposure data along with customer repayment records and their current payment activities. The proposed solution works best for predictive modeling operations which require credit default classification tasks.

The program performed a structural verification of the dataset through automated processes to confirm proper data loading and correct column labels and suitable data formats. The .xls file format requires special treatment because its data organization system needs correct management to prevent header position mistakes. import pandas as pd

```
 DATA_PATH = "default of credit card clients.xls"  df =

pd.read_excel(DATA_PATH, header=1, engine="xlrd")

print("Shape:", df.shape) print("Columns:", list(df.columns))

print(df.head(3)) print(df.dtypes)
```

Table 3.1

Variable Name	Description	Type	Role
ID	Unique customer identifier	Numeric	Identifier (Removed)
LIMIT_BAL	Credit limit assigned to customer	Numeric	Predictor
SEX	Gender (1 = Male, 2 = Female)	Categorical	Predictor
EDUCATION	Education level of customer	Categorical	Predictor
MARRIAGE	Marital status of customer	Categorical	Predictor
AGE	Age in years	Numeric	Predictor
PAY_1 – PAY_6	Repayment status for last 6 months	Numeric (Ordinal)	Predictor
BILL_AMT1 – BILL_AMT6	Monthly bill statement amounts	Numeric	Predictor
PAY_AMT1 – PAY_AMT6	Monthly payment amounts	Numeric	Predictor
DEFAULT	Default status next month (0 = No, 1 = Yes)	Binary	Target

Explanation:

The study uses this table to present all variables in an organized format which shows how identifiers and predictors relate to the target variable. The process of categorization functions as an essential step for preprocessing operations because it enables researchers to determine which variables require encoding and scaling.

3.3 Data Preprocessing

Data preprocessing was performed to ensure consistency, correctness, and suitability of the dataset for machine learning modeling. The dataset contained no missing data points or duplicate entries but needed multiple preprocessing steps to fix categorical code problems and establish uniform variable names and features which could be used for modeling.

3.3.1 Missing values and duplication

Before applying any transformation, the dataset was validated for completeness and record uniqueness. Missing values can distort statistical summaries and cause machine learning algorithms to fail or produce biased results, especially when the missingness is not random. Likewise, duplicate records can artificially amplify certain patterns, leading to overfitting and incorrect evaluation results. Therefore, the dataset was checked for missing values at the column level and duplicates at the row level as an initial data quality assurance step.

Code Snippet: Missing and Duplicate Checks

```
missing = df.isna().sum() print("Missing values:", missing[missing > 0])

duplicates = df.duplicated().sum()
print("Duplicate rows:", duplicates)
```

Explanation:

The function `df.isna().sum()` counts the number of null entries for each column. Sorting the results makes it easy to detect problematic features. The conditional output ensures the report can clearly state whether missing values exist. `df.duplicated().sum()` checks whether any entire rows are repeated. The results confirmed **no missing values** and **no duplicates**, indicating the dataset is already clean in terms of completeness, and no imputation or duplicate removal was required.

3.3.2 Target Variables Standardization

To ensure clarity throughout the preprocessing and modeling workflow, the original target variable name was standardized. The target variable in the raw dataset is labeled *default payment next month*, which is lengthy and can reduce code readability. Renaming it to **DEFAULT** improves consistency across the project, especially when multiple team members are sharing code and reporting outputs.

Additionally, the dataset may contain a known naming inconsistency where the most recent repayment status column is labeled PAY_0. In many references and publications, this same attribute is denoted as PAY_1 because it represents the first month in the six-month sequence used. Renaming PAY_0 to PAY_1 increases interpretability and avoids confusion when describing the repayment status variables.

Code Snippet

```
df = df.rename(columns={"default payment next month": "DEFAULT"})


if "PAY_0" in df.columns:

    df = df.rename(columns={"PAY_0": "PAY_1"})
```

Explanation:

This step does not alter the underlying data values; it only improves naming consistency. Standardizing variable names reduces future errors (e.g., incorrect column referencing) and allows the team to communicate results using a single agreed-upon notation.

3.3.3 Handling inconsistent categorical codes

Although the dataset contains no missing values, several categorical variables contain **invalid or undefined numeric codes**, which function similarly to “dirty data.” If such values are left unchanged, machine learning algorithms may treat them as legitimate categories. This creates unnecessary noise, increases the dimensionality of encoded features, and potentially lowers predictive performance. Therefore, categorical data cleaning was applied based on commonly accepted dataset documentation and standard preprocessing practices used in credit default prediction studies.

For **EDUCATION**, the expected categories are typically:
1 = Graduate School, 2 = University, 3 = High School, 4 = Others.
However, the raw dataset includes values **0, 5, and 6**, which do not correspond to defined categories. These were treated as “Others” and mapped to value 4.

For **MARRIAGE**, the expected categories are:
1 = Married, 2 = Single, 3 = Others.
However, the dataset includes value **0**, which is an invalid category. This was mapped to “Others” (value 3).

Code Snippet: Category Cleaning

```
def clean_education(x):
    return 4 if x in [0, 5, 6] else x

def clean_marriage(x):
    return 3 if x == 0 else x

df["EDUCATION"] = df["EDUCATION"].apply(clean_education) df["MARRIAGE"] =
df["MARRIAGE"].apply(clean_marriage)
```

Explanation: The cleaning functions explicitly correct invalid codes using deterministic rules, ensuring reproducibility. Reclassification into “Others” prevents the model from learning misleading relationships from undefined categories, while still retaining those records (no unnecessary deletion). This improves data integrity without reducing sample size.

Table3.2: Categorical Cleaning Rules

Variable	**Invalid Codes**	**Action Taken**	**Final Category**
EDUCATION	0, 5, 6	Reclassified	Others (4)
MARRIAGE	0	Reclassified	Others (3)
SEX	None	No change	Original

3.3.4 Removal of Non-Predictive Identifier

The dataset includes an `ID` column, which is a unique identifier assigned to each customer record. While it is useful for database indexing or record tracking, it does not represent customer behavior or financial risk. Including such an identifier in machine learning features may harm model training because models can mistakenly interpret ID patterns as predictive signals (especially for tree-based learners), resulting in spurious relationships and poor generalization.

Therefore, the `ID` column was removed from the set of predictive features to ensure that the models learn from meaningful variables only.

Code Snippet

```
drop_cols = ["ID"]  feature_cols = [c for c in df.columns if c not
in ["DEFAULT"]] feature_cols = [c for c in feature_cols if c not
in drop_cols]
```

Explanation:

This approach ensures that the dataset still retains the ID in the raw cleaned file (if needed for tracing), but it is not used during predictive modeling. This is considered best practice in supervised learning tasks.

3.3.5 Initial Exploratory Data Analysis: Target Distribution

Before splitting the dataset and applying transformations, an initial exploratory analysis was conducted to understand the distribution of the target variable. In credit default datasets, it is common for default cases to be less frequent than non-default cases. Such imbalance can lead to misleading evaluation if accuracy alone is used; a naive model could achieve high accuracy by always predicting the majority class.

Therefore, the counts of default and non-default cases were computed, and the default rate was calculated as a percentage. A bar chart was also generated to visually confirm the imbalance.

Code Snippet

```
target_counts = df["DEFAULT"].value_counts()
print(target_counts)

import matplotlib.pyplot as plt target_counts.plot(kind="bar")
plt.title("Default vs Non-Default Distribution")
plt.xlabel("DEFAULT") plt.ylabel("Count")
plt.show()
```

Explanation:

The output confirms that the dataset is imbalanced, with a significantly larger number of nondefault cases than default cases. This finding is important because it directly influences later modeling decisions such as the selection of evaluation metrics (precision, recall, F1-score, ROC-AUC) and potentially model strategies (class weighting or resampling).

3.3.6 Feature Grouping for Preprocessing

To apply correct preprocessing techniques, all predictors were divided into **categorical features** and **numerical features**. This separation is necessary because categorical variables should not be scaled like continuous variables, and numerical variables should not be one-hot encoded unnecessarily. Furthermore, treating categorical codes (e.g., education level) as numeric values could incorrectly impose ordinal meaning and create misleading distance relationships.

Code Snippet

```
categorical_cols = ["SEX", "EDUCATION", "MARRIAGE"]
numeric_cols = [c for c in feature_cols if c not in categorical_cols]
```

Explanation:
This grouping ensures that each feature receives the appropriate transformation: encoding for categorical variables and scaling for continuous variables.

Table 3.3: Feature Categorization

Category	Variables
Categorical	SEX, EDUCATION, MARRIAGE
Numerical	LIMIT_BAL, AGE, PAY_, *BILL_AMT*, PAY_AMT*
Target	DEFAULT

3.3.7 Train-test Split with Stratification

To evaluate models fairly, the dataset was split into training and testing sets using an 80/20 ratio. The split was performed using stratification based on the target variable DEFAULT. Stratification ensures that the proportion of default and non-default cases remains consistent in both training and test sets. Without stratification, the test set might contain disproportionately fewer default cases, causing unstable evaluation metrics.

Code Snippet

```
from sklearn.model_selection import train_test_split

X = df[feature_cols]
y = df["DEFAULT"]

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42, stratify=y
)
```

Explanation:

Using `random_state=42` ensures reproducibility. The `astype(int)` ensures that the target variable is correctly treated as an integer binary class. The split prepares the data for model training (training set) and unbiased model performance validation (test set).

3.3.8 Feauture Encoding and Scaling

In machine learning workflows, it is essential that all preprocessing steps applied to training data are consistently applied to test data using the same fitted parameters. If preprocessing is performed separately on training and test data, it results in **data leakage** or inconsistent feature representations. To prevent this, a reusable preprocessing pipeline was created using `ColumnTransformer`.

- Numerical attributes were standardized using `StandardScaler`, which centers values to mean 0 and scales them to unit variance.
- Categorical attributes were one-hot encoded using `OneHotEncoder`, which creates binary indicators for each category and avoids false ordinal assumptions.

The pipeline was fitted **only on the training set** to ensure that information from the test set is not used during training.

Code Snippet

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler

preprocessor = ColumnTransformer(
transformers=[
        ("num", StandardScaler(), numeric_cols),
        ("cat", OneHotEncoder(handle_unknown="ignore"), categorical_cols),
    ]
)

preprocessor.fit(X_train)
```

Explanation:

The argument `handle_unknown="ignore"` ensures robustness if unseen categories appear in future prediction data (or edge cases). Fitting only on the training set ensures proper machine learning methodology and improves the trustworthiness of evaluation results.

3.3.9 Exporting Cleaned Data and Pipeline

To support group collaboration and reproducibility, the cleaned dataset, train-test splits, and the fitted preprocessing object were exported into a dedicated folder. This allows other group members

(especially modeling and evaluation members) to use the same cleaned and standardized inputs without repeating preprocessing steps, ensuring consistent results across the report.

Code Snippet

```
import os, joblib

os.makedirs("member1_outputs", exist_ok=True)

df.to_csv("member1_outputs/data_cleaned.csv", index=False)
X_train.to_csv("member1_outputs/X_train.csv", index=False)
X_test.to_csv("member1_outputs/X_test.csv", index=False)
y_train.to_csv("member1_outputs/y_train.csv", index=False)
y_test.to_csv("member1_outputs/y_test.csv", index=False)

joblib.dump(preprocessor, "member1_outputs/preprocessor.pkl")
```

Table 3.4: Exported Files

File	Purpose
data_cleaned.csv	Clean dataset for EDA
X_train / X_test	Model training/testing
y_train / y_test	Labels
preprocessor.pkl	Reusable preprocessing pipeline

3.4 Data Mining

This step is aimed at constructing predictive models that may be used to categorize that a credit card client is likely to default the next month. Since the target variable (DEFAULT = 0 or 1) is binary, such a problem can be defined as a supervised binary classification task. Various machine learning models were used and tested against each other to select the most practical model to predict credit default.

To be consistent and avoid data leakage, all the models were combined with the preprocessing pipeline presented in Section 3.3.8. This will ensure that the feature scaling and categorical encoding are used in the same way with both training and testing data.

Chosen Classification Model(s).

Three popular classification algorithms were chosen because they complement each other as well as the applicability to financial risk modeling:

Logistic Regression

The reason why Logistic Regression can be used as a baseline model is because of its interpretability and effectiveness. It is a logistic-based model of probability of default and the predictors are assumed to be linearly related to the log-odds of the target variable. However, although simple, Logistic Regression is typically applied to the banking industry due to the availability of clear boundaries of the decision and clear interpretation of the coefficients.

Random Forest Classifier

Random Forest is a type of ensemble learning that builds several decision trees and consists of a majority vote on the predictions of these trees. This method causes less overfitting and is able to model complex and non-linear feature relationships. Random Forest can be employed especially in addressing high-dimensional data and the relationship between repayment history, billing amounts, and demographic characteristics.

Multilayer Perceptron (Neural Network).

The feedforward Neural Network was used to compute complicated non-linear trends in customer behavior. Neural Networks are better suited to represent some delicate relationships between repayment patterns, value of bills and time-dependent payment patterns. They are not as interpretable as the Logistic Regression, but they tend to have a better predictive performance in large datasets.

Model Implementation

All the models were trained on the same training data and preprocessing. The initial application of default hyperparameters was to create baseline performance and then hyperparameters were optimized by applying Grid Search.

Example code implementation structure:

```
from sklearn.pipeline import Pipeline

from sklearn.linear_model import LogisticRegression from

sklearn.ensemble import RandomForestClassifier from

sklearn.neural_network import MLPClassifier


models = {

   "Logistic Regression": LogisticRegression(max_iter=1000, class_weight="balanced"),

   "Random Forest": RandomForestClassifier(n_estimators=200, random_state=42),

   "Neural Network": MLPClassifier(hidden_layer_sizes=(64,32), max_iter=500, random_state=42)  }
```

The parameter class_weight="balanced" was applied where appropriate to address class imbalance, ensuring that default cases were not underrepresented during model training.

3.5 Evaluation

Modeling analysis is an important aspect of this research, especially because the data used is imbalanced and non default cases outnumber cases of default in a great ratio. It would be wrong to base it only on accuracy; thus, various performance measures were employed to present the overall evaluation.

Evaluation Metrics To measure each of the models, the following metrics were chosen,

Accuracy: This is a measure of the the overall accuracy of the predictions but it can be biased to the majority class.

Precision: Refers to the ratio of correct predictions on the total number of predictions.

Recall: Determines the capacity of the model to distinguish actual cases of default, which is important when managing risk.

F1-Score: The harmonic mean of precision and recall which is a balanced measure when there is an imbalance between the classes.

ROC-AUC (Receiver Operating Characteristic -Area Under Curve): Measures the model to differentiate between default and non-default classes at various probability levels.

Evaluation Procedure Each model was tested with the unseen test dataset after the training. Predictions were made and probability scores were drawn wherever possible so that ROC-AUC could be calculated.

Code implementation structure

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

y_pred = model.predict(X_test_transformed) y_prob =
model.predict_proba(X_test_transformed)[:,1]

accuracy = accuracy_score(y_test, y_pred) precision =
precision_score(y_test, y_pred) recall =
recall_score(y_test, y_pred) f1 = f1_score(y_test,
y_pred) roc_auc = roc_auc_score(y_test, y_prob)
```

Evaluation Rationale

Recall and F1-score were given particular emphasis because failing to detect high-risk customers (false negatives) can result in significant financial losses for banks. ROC-AUC was used to compare models independently of decision thresholds, offering a robust measure of overall discriminative power.

This multi-metric evaluation framework ensures that the selected model is not only accurate but also reliable and effective for identifying high-risk credit card clients in real-world financial decision-making scenarios

3.6 Results and Discussion

3.6.1 Demographic Risk Factors

3.6.1.1 Default Risk by Gender Analysis

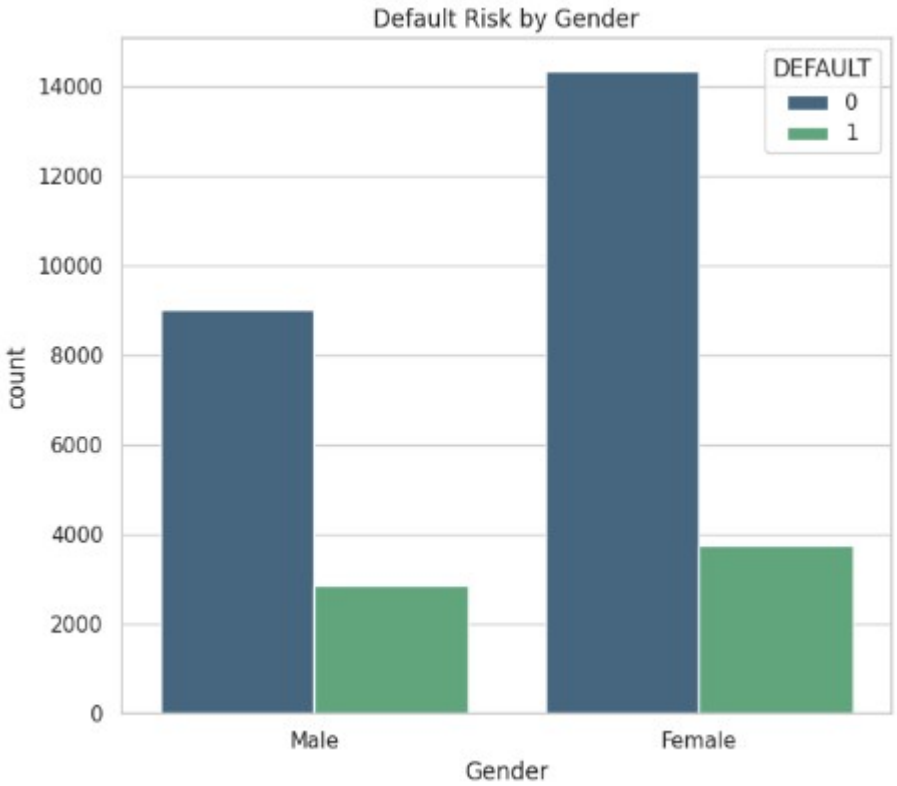


Figure 3.1: Distribution of Default Risk by Gender

Based on the Figure 3.1, the exploratory data analysis reveals that female clients constitute a significantly larger portion of the total customer base compared to male clients. The absolute number of default cases (shown by DEFAULT = 1) is significantly greater among females, totalling about 4,000 occurrences compared to roughly 3,000 for males, even though the majority of both groups are non-defaulters. This distribution highlights the report's emphasis on employing sophisticated data mining techniques to differentiate between dependable borrowers and those at danger of default across various social categories, confirming the existence of class imbalance within the demographic parameters.

3.6.1.2 Default Risk by Education Level Analysis

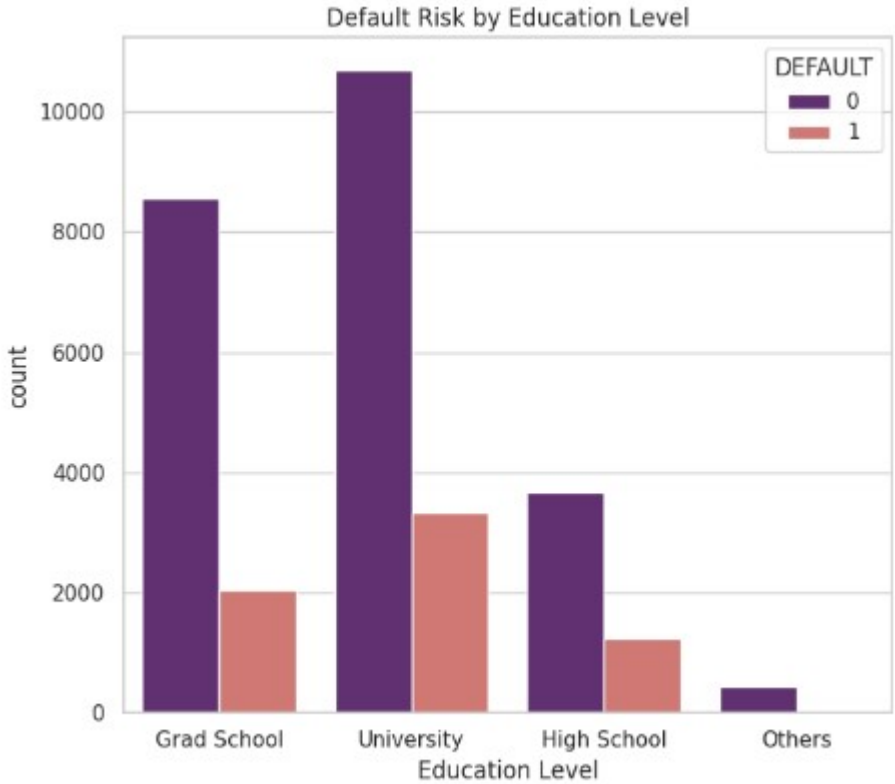


Figure 3.2: Distribution of Default Risk by Education Level

University-educated clients are the most common in the sample and account for the greatest number of defaults, according to Figure 3.2, followed by graduate school and high school graduates. These findings support the study's objective of comprehending the financial and demographic variables that affect credit card default and draw attention to the glaring class disparity between default (1) and non-default (0) cases, which calls for the application of specific machine learning metrics and preprocessing methods to guarantee precise risk assessment.

3.6.1.3 Default Risk by Marital Status Analysis

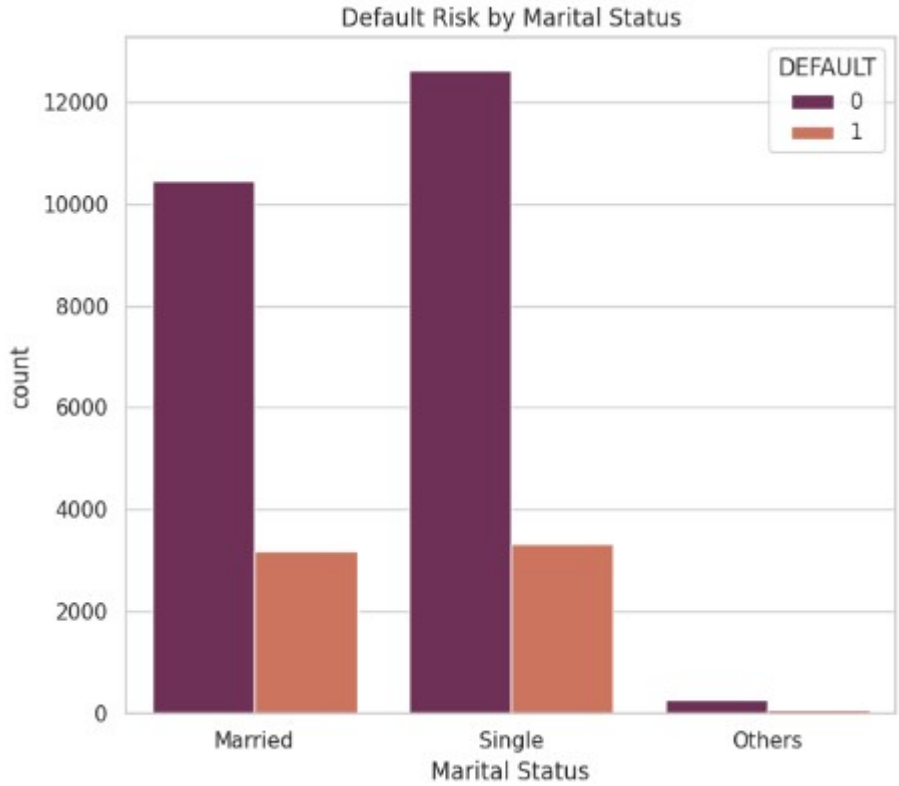


Figure 3.3: Distribution of Default Risk by Marital Status

Marital status data, as shown in Figure 3.3, reveals that married and single people account for the great majority of default occurrences, while the "Others" category which was cleansed during preprocessing remains a small portion. These results demonstrate the need for sophisticated data mining approaches like ensemble methods and feature scaling to enable reliable risk prediction, confirming the expected class imbalance where non-default instances (0) greatly outweigh default cases (1).

3.6.2 Credit Limit and Age Distribution

3.6.2.1 Credit Limit by Default Status Analysis

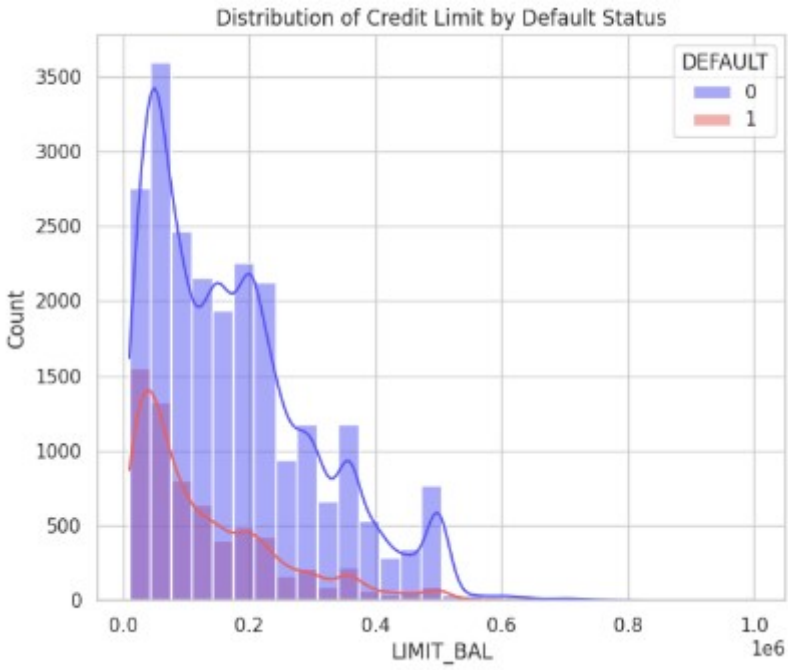


Figure 3.4: Distribution of Credit Limit by Default Status

Both default (1) and non-default (0) cases are concentrated at lower credit limits (LIMIT_BAL), with a clear decrease in frequency as credit limits rise (Figure 3.4). Together, these results show a glaring class imbalance where non-default cases much outnumber default cases, demonstrating the need for the study's methodology to correct data imbalances and employ metrics like ROC-AUC and F1-score for precise risk assessment.

3.6.2.2 Age Range Analysis

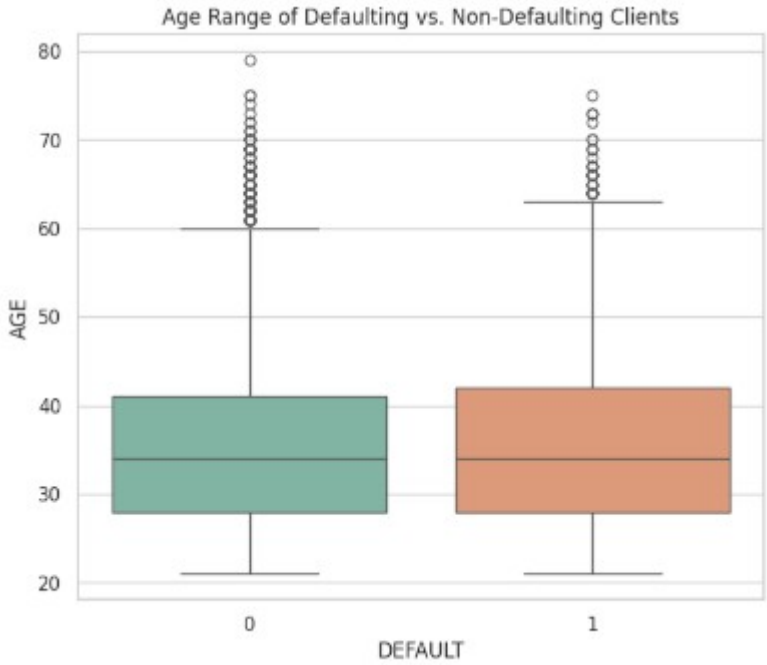


Figure 3.5: Age Range of Defaulting vs. Non-Defaulting Clients

The exploratory data analysis shows that the age distribution of defaulting and non-defaulting clients is strikingly comparable, as shown by the box plot in Figure 3.5. The median age of nondefaulters (0) and defaulters (1) is roughly 34 years, and the interquartile range (IQR) for both groups is generally between the late 20s and the early 40s. There are a lot of outliers in the nondefaulting group, with some clients getting close to 80 years old, even though the maximum age for frequent observations in both groups is around 60. This overlap implies that age may not be a reliable primary predictor of credit default on its own, supporting the report's methodology to assess it in conjunction with other financial and social characteristics including credit limits and repayment status to capture more nuanced consumer behaviours.

3.6.3 6-Month Repayment Trend vs. Default Risk

3.6.3.1 Repayment Status: PAY_1 Analysis

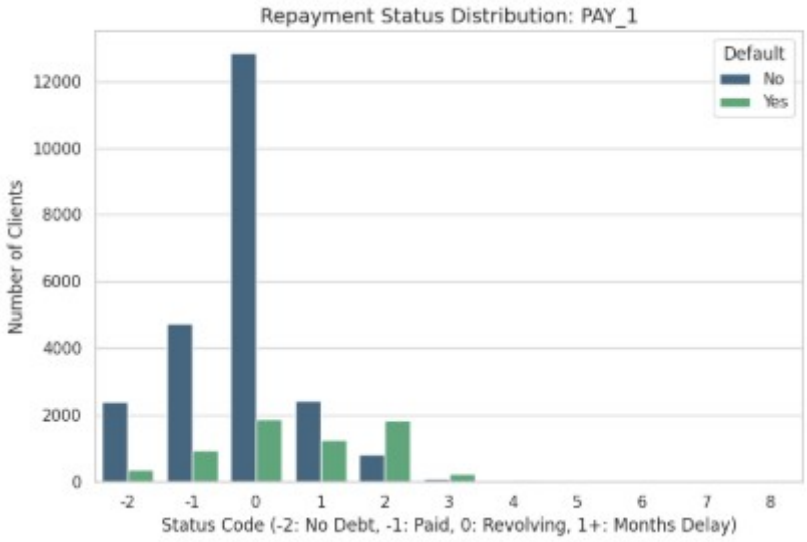


Figure 3.6: Repayment Status Distribution: PAY_1

According to Figure 3.6, the data shows that a sizable majority of clients are either in the negative categories (-1 and -2), which indicate that their payments are promptly paid or have no consumption, or in the category labelled 0, which denotes "use of revolving credit." On the other hand, a smaller but significant fraction of the dataset is spread among positive values (1 to 8), which represent the number of months that a payment has been past due. This distribution shows that although most clients retain a "on-time" status, there is a clear subgroup of the population displaying varied degrees of delinquency, which is an important predictor of the probability of default in the upcoming month.

Analysis

3.6.3.2 Repayment Status: PAY_2

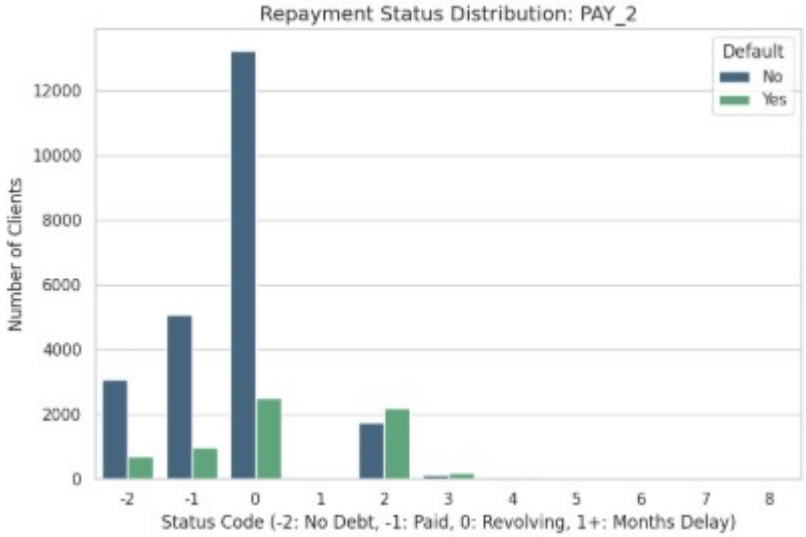


Figure 3.7: Repayment Status Distribution: PAY_2

The bulk of the 30,000 clients fall into the 0 (revolving credit), -1 (paid), or -2 (no debt) categories based on Figure 3.7, which is similar to Figure 3.6 and shows a relatively constant repayment behaviour for the majority of the population. This pattern is supported by the statistical summary, which displays a mean value of roughly -0.134, indicating that, on average, clients are either somewhat ahead of schedule or on track with their payments. Nevertheless, a portion of consumers with positive numbers between 1 and 8, which indicate the months of payment delay, are also included in the distribution. As a crucial historical signal for forecasting future defaults, this secondary peak or "tail" in the distribution reveals consumers who were already exhibiting delinquency a month before the most recent record.

Analysis

3.6.3.3 Repayment Status: PAY_3

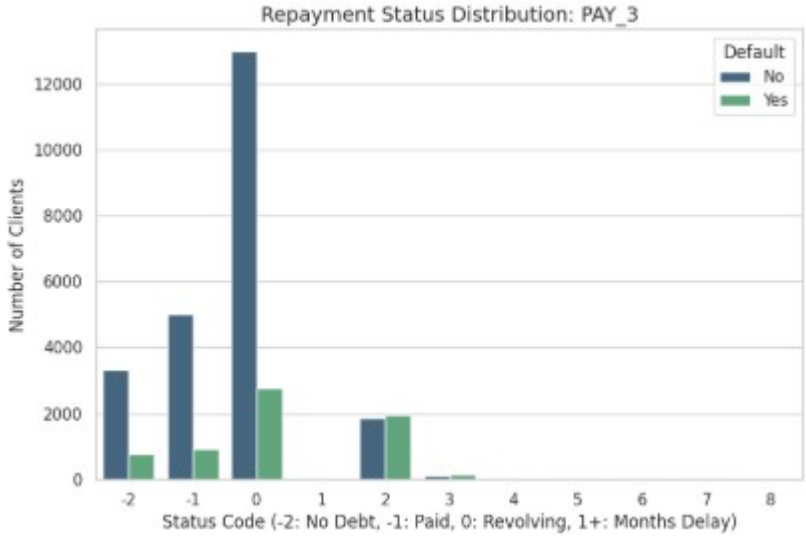


Figure 3.8: Repayment Status Distribution: PAY_3

The distribution in Figure 3.8 shows a similar pattern to prior months, with the great majority of clients falling into the -1 (paid), -2 (no debt), and 0 (revolving credit) categories. This suggests that most users are either not utilizing their credit line or are making timely payments. In particular, the statistical summary for PAY_3 reveals a mean value of almost -0.166 and a standard deviation of 1.197, indicating that, despite considerable fluctuation, the average behaviour tends somewhat toward early or on-time payment. There is a discernible "tail" of clients with values between 1 and 8, which reflect delays of one to eight months, even though most of the data is centred in these non-delayed categories. This PAY_3 historical repayment data is an essential component for Because it helps identify long-term trends of delinquency that distinguish between transient payment problems and ongoing default risk, the historical repayment data from PAY_3 is an essential component for the prediction models.

Analysis

3.6.3.4 Repayment Status: PAY_4

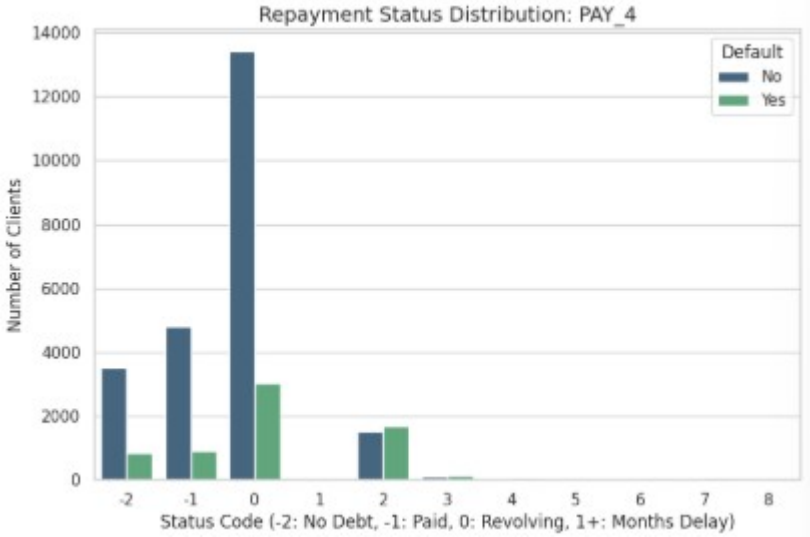


Figure 3.9: Repayment Status Distribution: PAY_4

Figure 3.9 shows that the distribution is significantly skewed toward the non-default categories, namely -2 (no debt), -1 (paid), and 0 (revolving credit). According to statistics, the mean for PAY_4 is roughly -0.221 with a standard deviation of 1.169, meaning that, on average, clients are somewhat more likely to make their payments on time or ahead of schedule than they were in previous months (PAY_1). The depiction shows a lower proportion of the population with delay status codes between 1 and 8, which indicate how many months a payment has been past due, despite this overall trend of stability. Because they offer crucial temporal context for detecting high-risk clients, these historical delay signals are crucial components for the machine learning models.

Analysis

3.6.3.5 Repayment Status: PAY_5

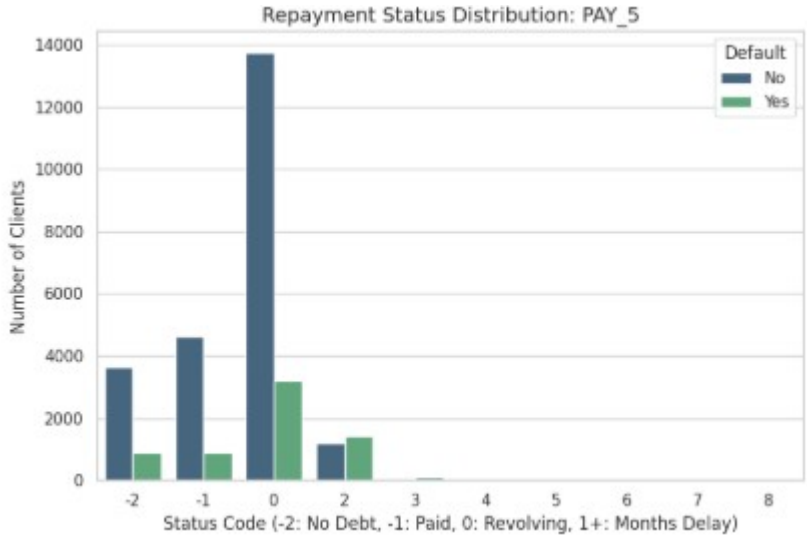


Figure 3.10: Repayment Status Distribution: PAY_5

Figure 3.10 shows that the distribution is still significantly skewed toward the non-default categories, with most clients falling into the -1 (paid), -2 (no debt), and 0 (revolving credit) classifications. According to statistics, PAY_5 has a mean value of roughly -0.266 and a standard deviation of 1.133, indicating that the customer population is generally inclined toward timely payments or inactive accounts during this time. A smaller portion of the population is spread throughout values 1 to 8, showing different durations of payment delays, even if most of the population is concentrated in these stable groups. Because it establishes the past payment patterns required to forecast the chance of default for the upcoming month, this repayment data is an essential component for the machine learning models.

Analysis

3.6.3.6 Repayment Status: PAY_6

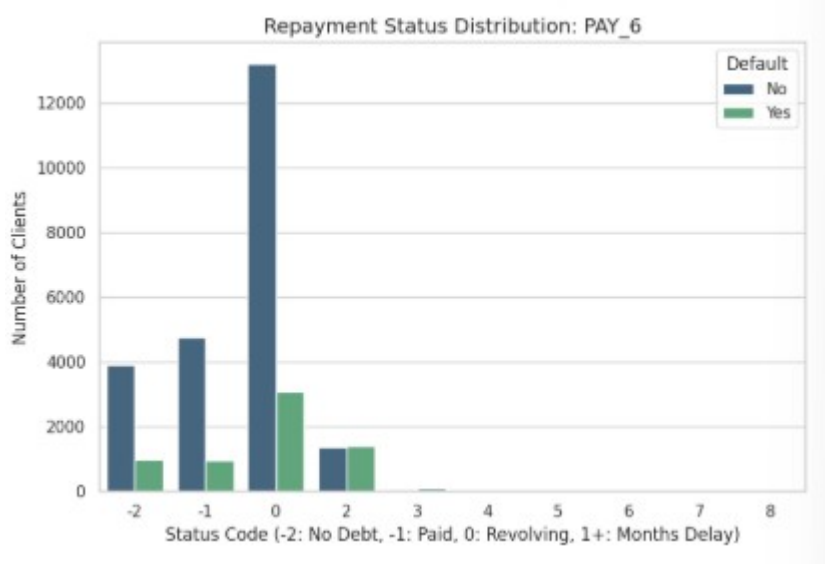


Figure 3.11: Repayment Status Distribution: PAY_6

According to Figure 3.11, the distribution exhibits a trend similar to that of the preceding months, with a significant concentration of clients in the non-default categories: -1 (paid), -2 (no debt), and 0 (revolving credit). With a mean of roughly -0.291 and a standard deviation of 1.150, PAY_6 had the lowest mean of the six-month period statistically, suggesting that clients were, on average, most likely to be up to date or ahead on their payments during this earliest reported month. A persistent "tail" of clients with delay status codes ranging from 1 to 8, which indicate the number of months a payment has been past due, is evident in the distribution despite this general stability. Because it preserves the oldest indicators of credit risk in the dataset, this historical baseline is essential for the predictive modelling.

Analysis

3.6.4 Bill Amount Trends (BILL_AMT1 to BILL_AMT6)

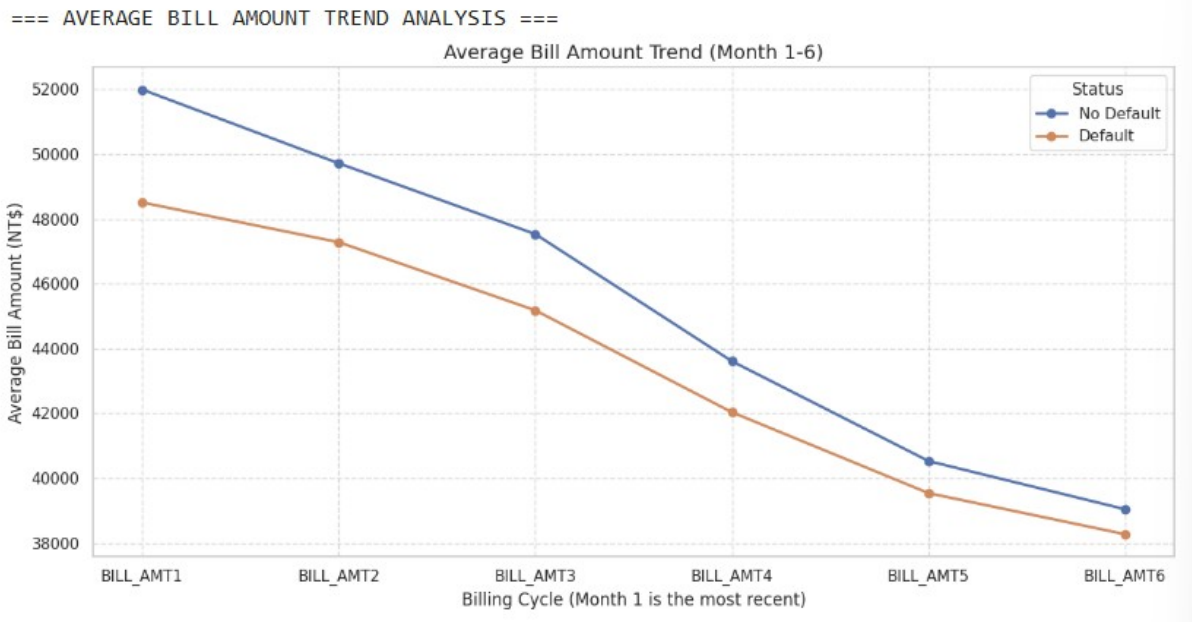


Figure 3.12: Average Bill Amount Trend (Month 1-6) Distribution

According to Figure 3.12, the statistical summary shows a general downward tendency in the average bill amount as we go back in time. The average starts at roughly 51,223 in the most recent month (BILL_AMT1) and progressively drops to 38,872 by the sixth month (BILL_AMT6). This pattern points to a consistent rise in credit use or spending within the time frame under observation. The median bill amounts follow a similar declining pattern from 22,381 to 17,071, confirming that the overall central tendency of billing was higher in the months closer to the default prediction target, despite the high standard deviations (ranging from ~59,554 to ~73,635), which show significant variance in individual spending habits. Debt levels could be a warning of an approaching default.

3.6.5 Payment Behaviour Analysis (PAY_AMT1 to PAY_AMT6)

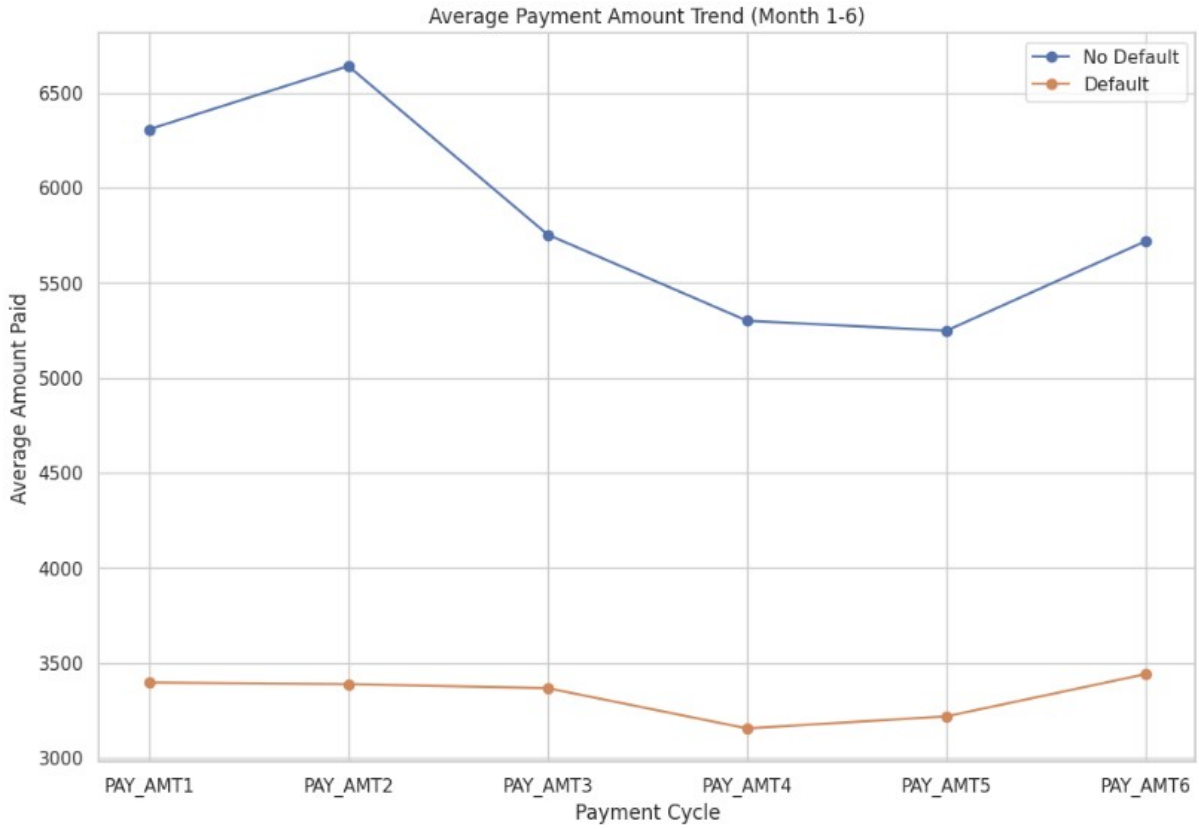


Figure 3.13: Average Payment Amount (Month 1-6) Distribution

Figure 3.13 shows that the average payment amount was comparatively constant over the course of the six months, with mean values ranging from roughly 4,799 to 5,921. However, as shown by maximum payments as high as 1,684,259, the standard deviations are remarkably high (reaching over 23,000 for PAY_AMT2), indicating extraordinary volatility in payment quantities and the presence of substantial outliers. It's interesting to see that the median (50th percentile) payments are consistently between 1,500 and 2,100, while the averages stay constant. This implies that the majority of people normally pay a much smaller, conventional amount each month, while a small percentage of clients make very significant payments. These payment amounts are critical features for the predictive models, as they provide insight into whether a client's payment volume is sufficient to cover their mounting bill amounts.

3.6.6 Correlation Heatmap of Credit Card Features Analysis

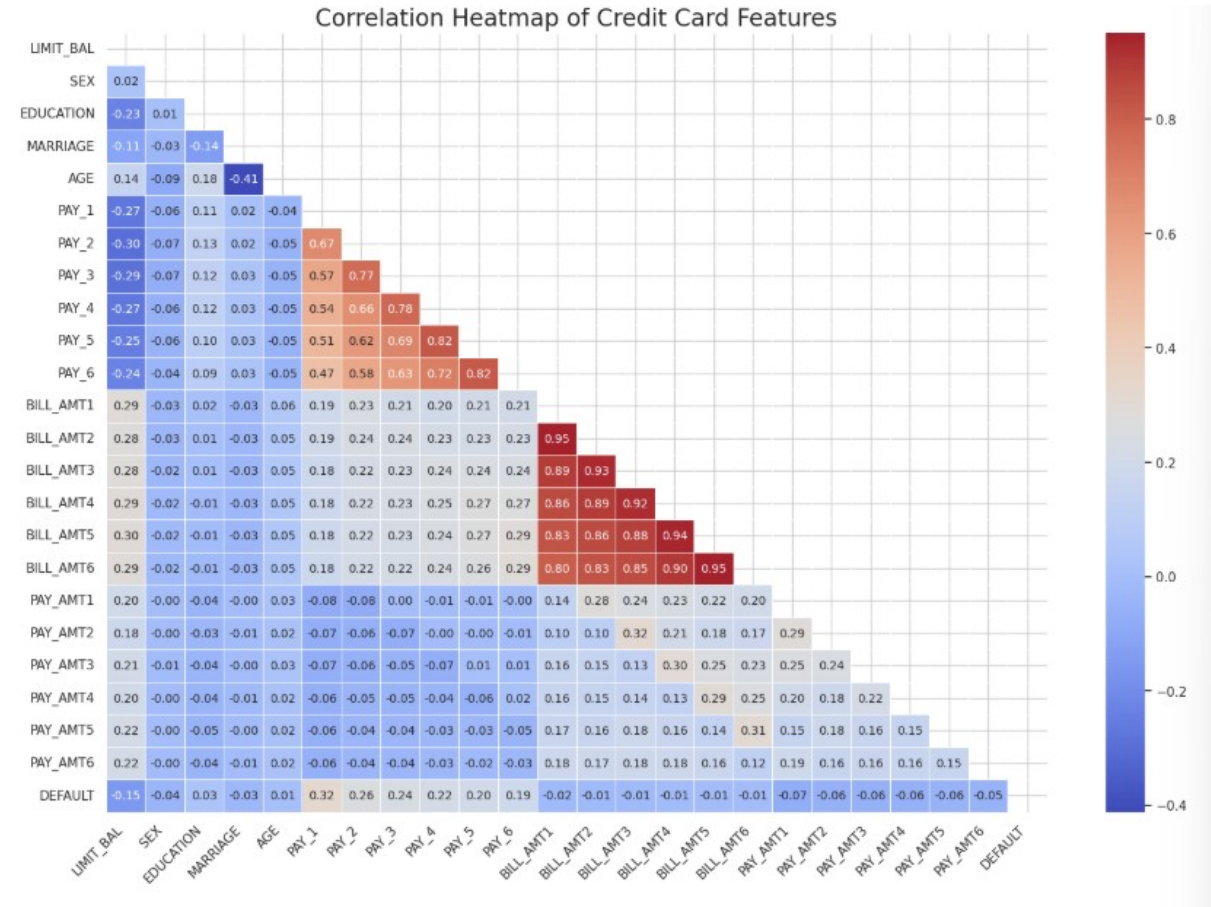


Figure 3.14: Correlation Heatmap of Credit Card Features

A thorough summary of the linear correlations between every item in the dataset, including credit limit, demographics, repayment history, billing amounts, and the target default variable, can be found in Figure 3.14. The billing amount variables (BILL_AMT1 to BILL_AMT6) show a remarkably significant positive association in the heatmap, suggesting that a client's statement balance in one month is a reliable indicator of their balance in the subsequent month. The payback status variables (PAY_1 to PAY_6) also show a strong positive connection, indicating that clients who have payment delays in one period are likely to repeat that trend in later months. It's interesting to note that the repayment status indicators, especially the most recent month (PAY_1), have the most direct link with the likelihood of default, whereas the demographic characteristics (AGE, SEX, and EDUCATION) have a relatively poor correlation with the DEFAULT goal. The feature selection and engineering strategies for the machine learning models are informed by the identification of a variety of features, like the bill amounts, which is made possible by this visual summary.

3.6.7 Evaluation Models in Data Mining

	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.686000	0.367696	0.583271	0.451049	0.690984
Random Forest	0.779333	0.501153	0.491334	0.496195	0.742538
Neural Network	0.729833	0.414832	0.539563	0.469047	0.720523
Logistic Regression (Tuned)	0.778833	0.000000	0.000000	0.000000	0.635839
Random Forest (Tuned)	0.816000	0.652949	0.358704	0.463035	0.762706
Neural Network (Tuned)	0.773167	0.470486	0.204220	0.284813	0.651509

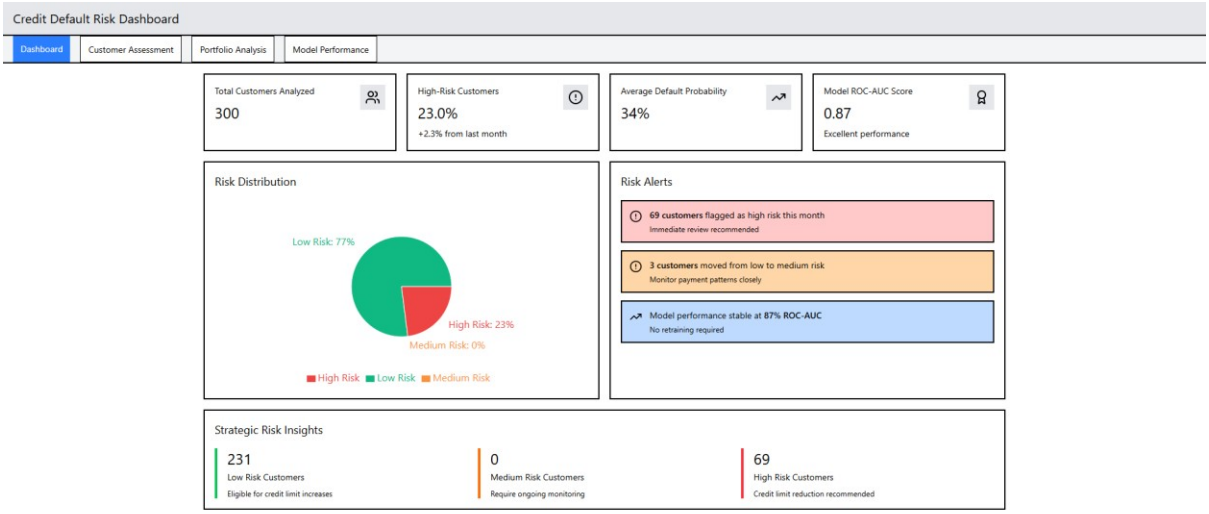
Table 3.1: Comparison of Data Mining Models

The original test set was kept for objective assessment. However, the Synthetic Minority Oversampling Technique (SMOTE) was only used to the training data in order to solve the significant class imbalance in the credit card default dataset. To offer a fair assessment of the efficacy of classification, model performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. With an accuracy of roughly 0.82 and the greatest ROC-AUC of roughly 0.76, the tuned Random Forest model showed the best overall performance, demonstrating greater discriminatory power between default and non-default clients. Furthermore, the model's capacity to accurately detect default instances while managing false positives was demonstrated by its comparatively balanced precision and recall. Despite its capacity to capture non-linear interactions, the neural network model's recall shown notable variability after tuning, indicating sensitivity to parameter settings. It also attained reasonable accuracy and ROC-AUC values. The modified Logistic Regression model, on the other hand, showed a significant drop in recall and F1-score and was unable to accurately categorize default instances even after SMOTE application. This underscores the model's limits in simulating intricate financial patterns in unbalanced datasets. The tuned Random Forest model is the best option for real-world credit default risk assessment since it provides the most dependable trade-off between predictive accuracy, minority-class recognition, and resilience, according to the descriptive performance criteria.

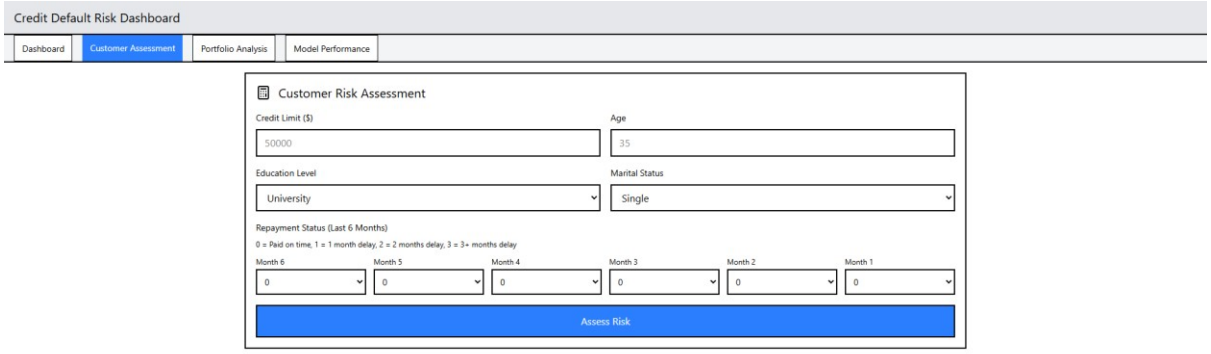
3.7 Application

In order to show how customers can utilise the predictive model for credit default risk assessment, an application mockup was created. This application mockup can help customers effectively interact with the credit default prediction model by translating complex predictive outputs into clear visual and actionable insights, the application enhances the usability and practical value of the underlying model.

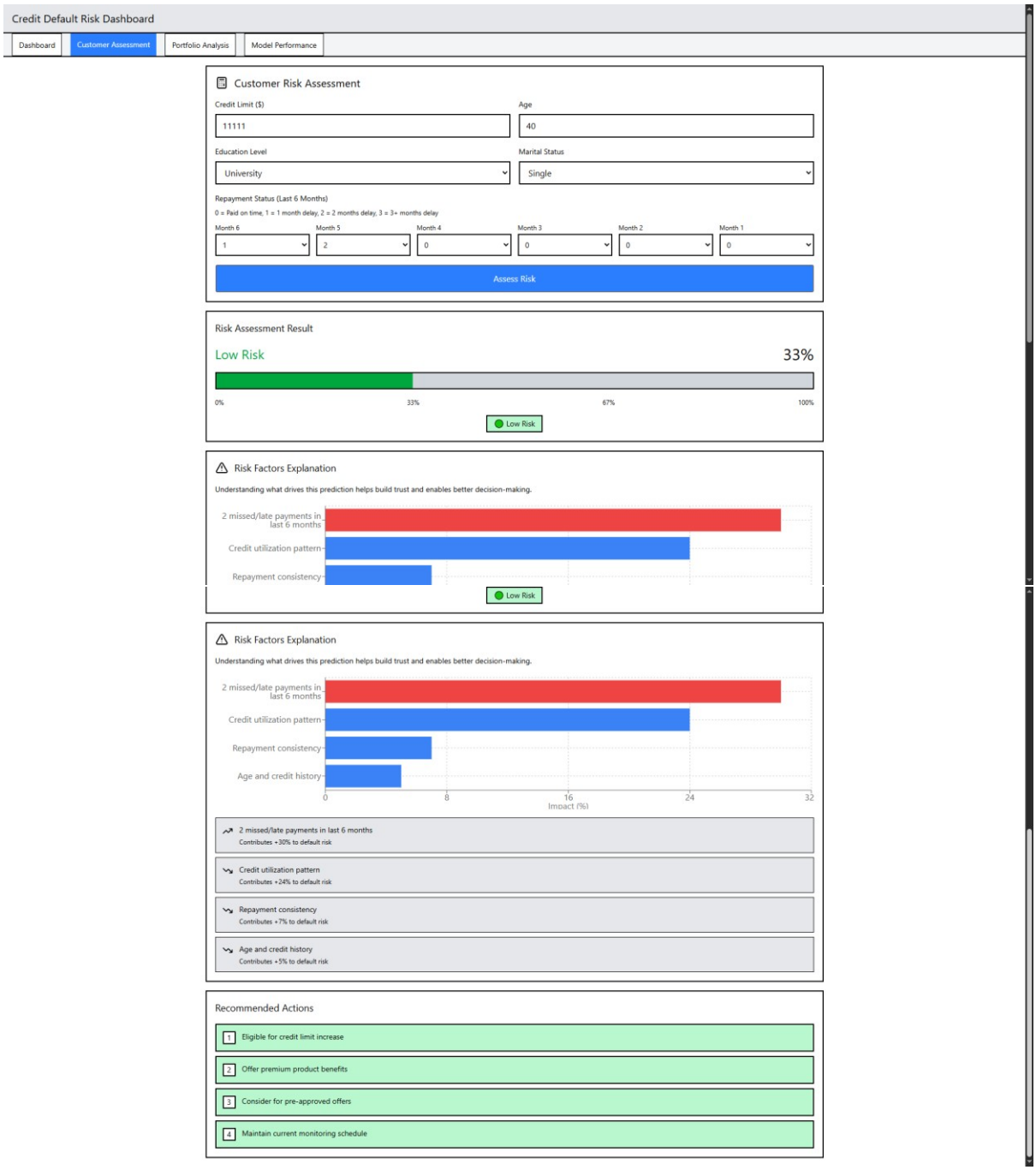
First of all, we got the main dashboard. The main dashboard visualises the customer's data into 1 single page. On the top of the page we can see the dashboard name and several navigation buttons which leads to their respective pages. The main dashboard shows the general data visualisation of the dataset, which includes the total number of customers analysed, alongside showing the percentage of high risk customers, derived from the total customers, average default property and the Model ROC-AUC Score. The dashboard also presents a risk distribution pie chart alongside risk alerts for the users of the dashboard to take note of. Finally, the strategic risk insights allows the users to determine if a customer should be allowed for a credit raise, monitor medium risk customers or limit customer's credit limit.



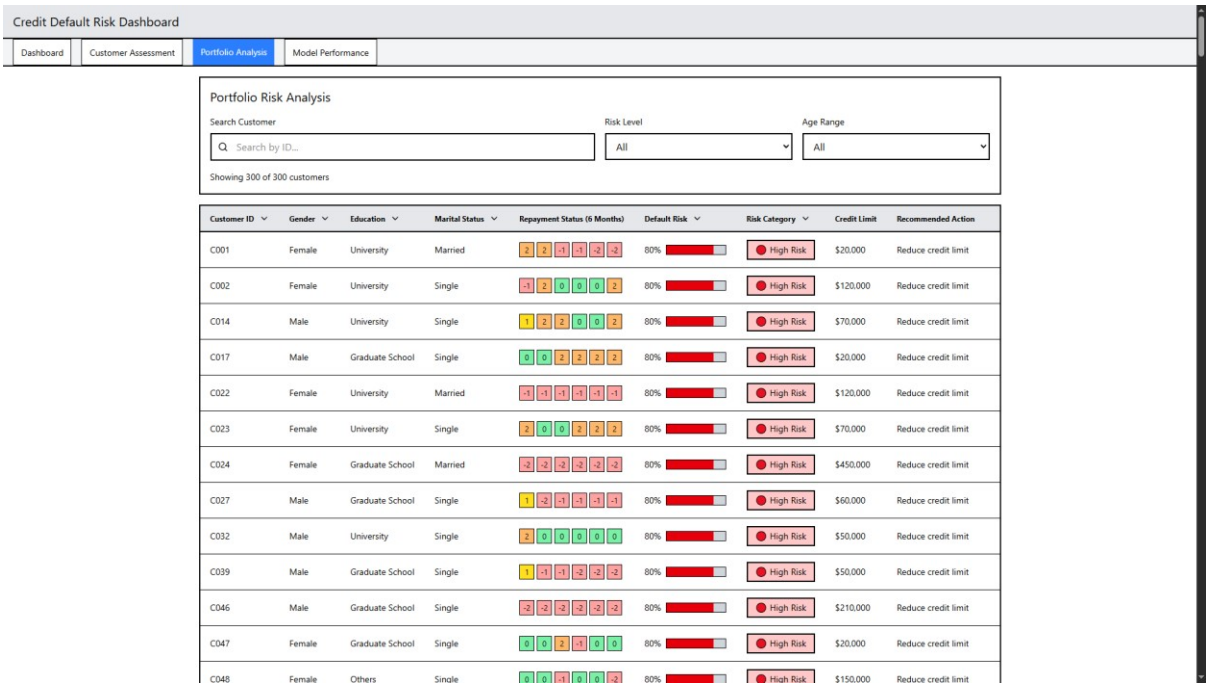
Next, we have a customer risk assessment. Essentially, it is just a risk calculator. Users input custom values to determine a customer's risk.



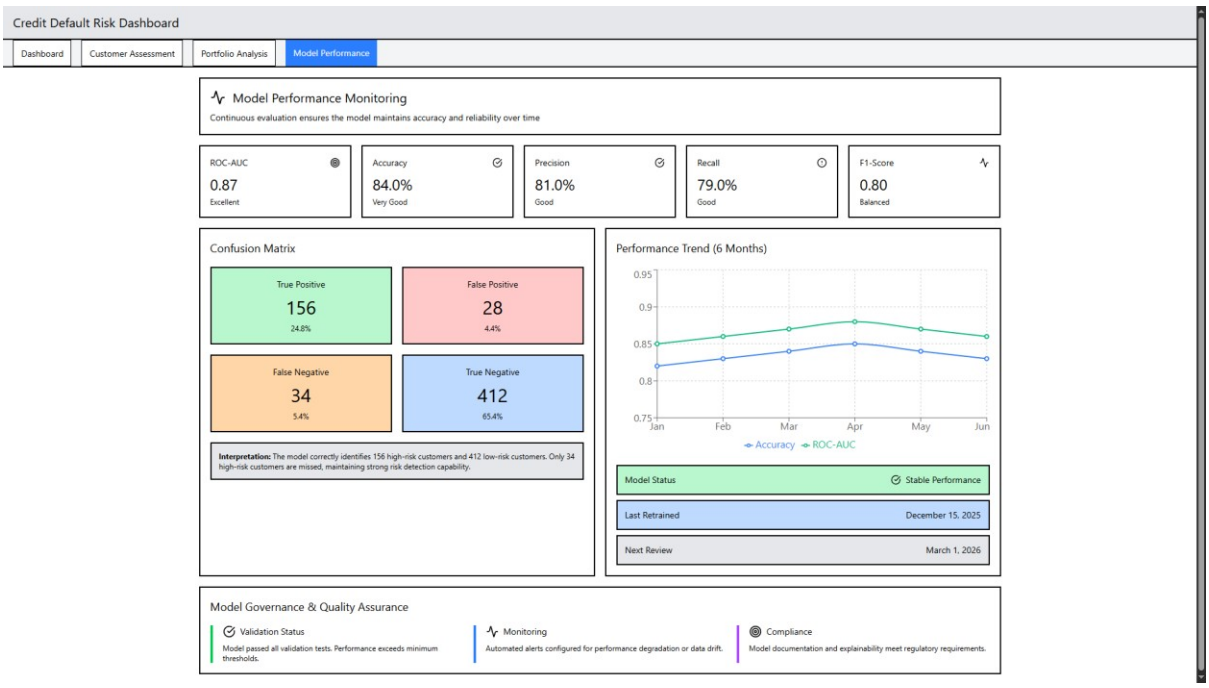
After inputting the needed values and clicking on the assess risk button, the results will be displayed, determining the customer's risk level, provides reason for the risk level, and recommended actions for the customer.



Next, we have the portfolio analysis page, where the list of customers and values can be seen. Users can filter for certain results by clicking on the top rows or searching via the search bar.



Finally, we have the model performance page, showing the model's performance and ensuring that the model is up to date and ready to deploy.



Model Performance Evaluation

The credit default prediction model was evaluated using a set of widely adopted classification performance indicators, namely **accuracy, precision, recall, F1-score, and the area under the Receiver Operating Characteristic curve (ROC-AUC)**. These metrics collectively provide a comprehensive assessment of the model's predictive effectiveness, particularly in the context of credit risk analysis where class imbalance is commonly observed.

Accuracy

The model achieved an **accuracy of 0.84**, indicating that 84% of all observations were correctly classified. This result demonstrates a strong overall predictive capability. However, accuracy alone is not sufficient to fully evaluate credit risk models, as it does not account for the differing costs associated with misclassification errors, particularly in identifying high-risk customers.

Precision

The **precision score of 0.81** reflects the model's ability to correctly identify customers who are likely to default. This indicates that 81% of customers predicted to be defaulters were indeed actual defaulters. High precision is particularly important in financial decision-making, as it minimizes the misclassification of low-risk customers as high-risk, thereby reducing unnecessary credit restrictions.

Recall

The model obtained a **recall value of 0.79**, demonstrating that 79% of actual default cases were correctly detected. This highlights the model's effectiveness in identifying high-risk customers. A strong recall is essential in credit risk assessment, as undetected defaulters can lead to significant financial losses for lending institutions.

F1-Score

The **F1-score of 0.80** indicates a balanced trade-off between precision and recall. This metric confirms that the model maintains consistency in identifying default cases while simultaneously controlling false positive rates. The F1-score is particularly useful in evaluating model performance when both false positives and false negatives carry meaningful consequences.

ROC-AUC

The model achieved a **ROC-AUC score of 0.87**, indicating excellent discriminative capability between default and non-default customers. A ROC-AUC value approaching 1.0 suggests that the model is highly effective at ranking customers according to their likelihood of default across varying classification thresholds.

Overall Evaluation

In summary, the model demonstrates **robust and reliable performance** across all evaluated metrics. The high ROC-AUC score, combined with balanced precision and recall values, suggests that the model is well-suited for credit default risk assessment. These results indicate that the model can serve as an effective decision-support tool for financial institutions, enabling improved credit risk management and informed lending decisions.

4.Conclusion

Using data mining techniques on a dataset of 30,000 Taiwanese clients, this work effectively created a methodology for predicting credit card default risk. Several machine learning models were assessed using a strict methodology that included feature scaling, data preparation, and using methods like SMOTE to address class imbalances. With an accuracy of 0.82 and a ROCAUC of 0.76, the Random Forest model outperformed Logistic Regression and Neural Networks in identifying high-risk clients.

This work's potential applications as a decision-support tool for financial institutions is what makes it practically significant. Bank executives can evaluate default probability and obtain strategic insights that can be used to make decisions about credit limit modifications or to monitor medium-risk portfolios in real-time by incorporating the predictive model into a dashboard prototype. In a world economy that depends more and more on credit, such a technology is essential for reducing financial losses and streamlining credit approval procedures.

Future directions for this work could focus on overcoming current limitations and exploring new dimensions of the data:

- Addressing Data Sensitivity: Although the Neural Network had promise, parameter adjustments had a significant impact on its performance. To stabilize results, future rounds might investigate deeper learning structures or more sophisticated hyperparameter tuning.
- Temporal and Real-Time Data: A static dataset is used in this investigation. The model's ability to react to abrupt changes in the market may be improved by using real-time transaction data and more comprehensive economic indicators.
- Feature Expansion: According to exploratory investigation, age might not be a reliable indicator on its own. In order to further improve risk profiles, future research could investigate manufactured elements that integrate demographic information with more intricate behavioral patterns, including spending-to-income ratios or psychometric credit scoring.
- Model Explainability: Further study into "Explainable AI" (XAI) may enable bank managers comprehend the precise "why" behind a high-risk score, guaranteeing transparency and regulatory compliance in automated financial choices, even though Random Forest is an excellent model.

5.References

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https://d1wqtxts1xzle7.cloudfront.net/120173870/Model_for_the_Prediction_of_Default_Risk_of_Funding_Requests_Using_Data_Mining_Sameh_Ali_2-libre.pdf?1734018356=&responsecontentdisposition=inline%3B+filename%3DModel_for_the_Prediction_of_Default_Risk.pdf&Expires=1767158594&Signature=Gdinwsc0VXnPp47cmhOxz1i9OuG5Q9EeD7MLZg7YiqwUDYARZgPveq~ ZB4-OzGaynfBJnkYBO4CPoaDDJGeybHA~C0suMwVf10z27cOZgY2u2IZiKfX4irbblA9jkQLFB1zF1tyK5F3aGdDHZbRJcF~XnpuqcD55nil73BxzargUwyKcx2Vd72ZqJx3oWZKdVc2C2~xtKe9BEc8se7BcAb~aIZQA0WtB26aDMLDTKiMNfRfhMq ~sJ8flu0-Z6gzTAg31VPgBSZqeyl8j0Da5VFtftylzWrIeb1GTZ~OTL~MzoPvWVsULmjCqZo087ipDZn8~uckOlV7xCdTvlFpw__&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA

6.Appendix

1. https://laurel-food-52616486.figma.site/portfolio (Application Mockup link)